DHRUV DIAGNOSTICS, NAGPUR · VETERINARY VERTICAL

Veterinary testing on our existing analysers

Calibration, verification and validation plan for the ARCHITECT, MAGLUMI, Mindray BC-700 series CBC, BioSystems BA200 and the real-time PCR laboratory.

Prepared 15 October 2026 for the laboratory director and quality manager. Technical planning document; not legal advice.



1. Summary

None of our analysers needs a different **calibration** to accept animal samples: calibrators are traceable to human reference materials and stay on the manufacturers' schedules. What changes is **verification and validation**. Each test must be shown to work in each species' blood, on our method, and be reported against reference intervals set on that method. Some tests cannot be done on human platforms at all and need a veterinary reagent, a veterinary analyser or a partner laboratory.

System	Readiness	What to do
BioSystems BA200 (chemistry)	Ready with work	BioSystems markets the BA200 for veterinary use, with 120+ species-filtered reagents and veterinary reference materials. Load the veterinary applications, verify in canine and feline pools, set species reference intervals. Add DGGR lipase, fructosamine and bile acids.
Mindray BC-700 series (human CBC + ESR)	Limited	Usable for canine totals (RBC, HGB, HCT, WBC) after method comparison, with a blood-film differential on every animal CBC. Not reliable for feline platelets or nucleated-RBC species. Strongly consider a veterinary analyser (Mindray BC-60R Vet or BC-5000 Vet).
Abbott ARCHITECT (hormones)	Validate first	No published dog or cat validation of ARCHITECT hormone assays was found. Validate total T4, cortisol and progesterone in-house before launch. Canine TSH and free T4 by equilibrium dialysis cannot be done here.
Snibe MAGLUMI (hormones)	Validate first	No Snibe veterinary reagents and no published animal validation found. Run the same validation; keep whichever platform performs better per analyte, the other as backup.
Real-time PCR lab	Ready with work	Run commercial veterinary qPCR kits (tick fever, parvovirus, distemper, leptospira, bovine haemoparasites, mastitis) after kit verification. **Do not** offer avian influenza testing; human H1N1/H3N2 assays are not for animals.

Before any validation work: engage a registered veterinary pathologist (MVSc) to sign and interpret veterinary reports, and set up a species field and separate reference intervals in the LIS. Veterinary tests will run **outside the NABL scope** at launch: keep the NABL symbol off veterinary reports and say so on each report.

2. Calibration, verification and validation

- **Calibration** sets the instrument's response with the manufacturer's calibrators. Keep each system's calibration schedule unchanged (new reagent lot, interval in the instructions for use, after major maintenance). Do not recalibrate with animal material.
- **Verification** shows that a veterinary-labelled method performs as its maker claims in our hands (BA200 veterinary reagents, veterinary PCR kits, a veterinary haematology analyser).
- **Validation** shows that a method not designed for animals gives clinically usable results in a given species (human hormone immunoassays, the human Mindray analyser). ASVCP guidelines say this must be done before a new test or instrument is used.

Why human assays need species validation. In dogs, cats and horses, chemiluminescent total T4 read about 30-40% lower than radioimmunoassay, a bias not seen in human samples (Singh 1997). Immulite 2000 total T4 gave a slope of 0.76 in dogs and 0.71 in cats against a reference method (Wolff 2020). Canine progesterone by chemiluminescence read significantly higher than by radioimmunoassay three days before ovulation, which changes breeding decisions. Results are therefore **method-specific**: reference intervals and decision limits must come from our own method.

Standard validation experiments

Experiment	Design	Acceptance
Precision	Two pools of species serum or blood (low and high), 5 replicates a day for 5 days (CLSI EP15 design)	Within-laboratory CV small enough that total observed error meets the target below
Method comparison	At least 40 samples per species across the reportable range, split with a veterinary reference laboratory or validated veterinary analyser (CLSI EP09 approach); Passing-Bablok and Bland-Altman	Total observed error (|bias| + 2 CV) within ASVCP total allowable error (TEa) where published (Harr 2013 for canine, feline, equine biochemistry), otherwise a laboratory-set target
Linearity and dilution parallelism	High animal pool diluted in 5 steps with the assay diluent	Recovery 80-120% of expected at each step; for immunoassays, dilution curve parallel to the calibration curve
Spike recovery (immunoassays)	Known analyte added to a low animal pool at 2-3 levels	Recovery 80-120%
Low-end performance	Low pool run 10 times	CV of 20% or less at the clinical decision level (low T4 in hypothyroid dogs, cortisol after dexamethasone)
Interference	Spiking or paired samples with haemolysis, lipaemia and icterus	Set rejection or comment thresholds by index
Transport stability	Test at 0, 24 and 48 h at 4 °C	Define the maximum time for district samples

3. Instrument plans

3.1 BioSystems BA200: veterinary chemistry

Our BA200 is made by BioSystems S.A. (Spain). BioSystems markets the BA200 for veterinary analysis. Its veterinary reagent range covers more than 120 parameters, filterable by species (canine, feline, bovine, camel, equine, poultry, reptiles and exotics), and it lists veterinary reference materials (VetBC). The BA200 has 88 reagent and sample positions and an optional ISE for sodium, potassium and chloride.

- **Calibration:** use the calibrator named in each veterinary reagent's instructions; same schedule as human tests.
- **Verification:** precision and method comparison in canine and feline pools for each analyte; bovine pools for the dairy profile.
- **Quality control:** veterinary control material at two levels on every run, plus a pooled in-house species serum.
- **Albumin:** bromocresol green (BCG) reads higher than electrophoresis in dogs and horses, lower in ruminants and chickens, and higher still in canine heparin plasma because of fibrinogen. Use serum, report against method-specific species intervals.
- **Pancreatitis:** Spec cPL is an IDEXX test. DGGR lipase correlated closely with it in 301 dogs (rs 0.91; kappa 0.87 at 143 U/L for cPL above 200 ug/L). Offer DGGR lipase and set our own cut-off.
- **Add to the menu:** fructosamine, bile acids, CK, GGT, phosphorus, magnesium, beta-hydroxybutyrate (dairy ketosis), urine protein and creatinine for the UPC ratio.
- **Not possible:** SDMA (the validated immunoassay is an IDEXX product). Offer creatinine and UPC, and send SDMA to a partner if needed.

3.2 Mindray BC-700 series: haematology

Our analyser is a BC-700 series human model (BC-720, BC-760 or BC-780). Mindray describes the series as a combined CBC and ESR analyser on its SF Cube fluorescence platform, with a 5-part differential plus immature granulocytes and nucleated red cells (hence "7-part"), reticulocytes on the BC-780, and a platelet count that combines impedance and optical channels (PLT-H). No veterinary or species mode was found for this series.

Known limits of human analysers with animal blood: feline platelets overlap red cells in size, so most analysers cannot count them; platelet clumps (common in cats) give falsely low counts, which should be reported as minimums; birds, reptiles and fish have nucleated red cells and cannot be analysed automatically; some mammals have red cells small enough to be confused with platelets; the white-cell differential is gated for human cells.

- **If we use the human analyser:** validate RBC, HGB, HCT, MCV and WBC for dogs by comparison with a veterinary analyser on at least 40 samples; perform a 100-cell blood-film differential on every animal CBC; estimate feline platelets from the film; review the film for parasites (Babesia, Ehrlichia morulae, Hepatozoon).
- **Recommended:** add a veterinary analyser. Mindray's animal-health company sells the BC-60R Vet (6-part differential, 33 parameters, scatter-plus-fluorescence, about 40 samples an hour, feline-friendly, many species) and the BC-5000 Vet (5-part, 13 built-in species including dog, cat, horse, cow, goat and sheep). A veterinary analyser needs only verification, not full validation.
- **BC-700 specifics:** do not report the ESR for animals; the module is set up for human blood. If our unit is a BC-780, validate the canine reticulocyte count, which separates regenerative from non-regenerative anaemia (common in tick fever). PLT-H may cope better with large platelets, but feline platelets still need a smear estimate unless our own comparison shows otherwise.
- **Sample timing:** canine EDTA platelet counts begin to fall by 12 h and MCV rises within 12-24 h; ask clinics for a fresh smear with every EDTA tube.

3.3 Abbott ARCHITECT: hormones

We found no published validation of ARCHITECT hormone immunoassays (total T4, free T4, cortisol, progesterone, insulin) in dogs or cats. This is a negative search result, not proof that none exists; ask Abbott India for any veterinary data. On the chemistry side, a canine-specific CRP reagent (Gentian) has been validated on the ARCHITECT c4000 (Hillstrom 2014), which matters only if we have a c-series module.

Assay	Plan	Validation specifics
Total T4 (dog, cat)	Validate	Full experiment set; method comparison with a veterinary reference laboratory on 40+ canine and 40+ feline samples. Expect a proportional bias; set method-specific intervals and a grey zone.
Cortisol (dog)	Validate	Full set, plus dynamic testing: ACTH stimulation and low-dose dexamethasone suppression in at least 10 healthy dogs to confirm expected rises and suppression (the design used to validate Immulite cortisol).
Progesterone (bitch)	Validate	Method comparison against a reference method on serial samples through oestrus; derive our own ovulation and breeding thresholds. Do not use thresholds from another method.
Insulin	Not at launch	Species cross-reactivity unknown; defer.
Free T4	Send out	Analogue free T4 methods are unreliable in dogs; the reference method is equilibrium dialysis.
Canine TSH	Send out	Needs a canine-specific assay. Commercial canine TSH ELISA kits exist but those found are research-use only.

3.4 Snibe MAGLUMI: hormones

Snibe lists no veterinary reagent line or veterinary analyser, and no peer-reviewed MAGLUMI validation in any animal species was found. Run the same T4, cortisol and progesterone validation on the MAGLUMI in parallel with the ARCHITECT, using the same samples. For each analyte, keep the platform with lower bias against the reference laboratory and better low-end precision as the veterinary method, and the other as a documented backup.

3.5 Real-time PCR laboratory

Avian influenza: do not test. Avian influenza is a scheduled disease under the Prevention and Control of Infectious and Contagious Diseases in Animals Act, 2009. ICAR-NIHSAD Bhopal is designated to confirm any case in India. Our H1N1 and H3N2 assays are human diagnostics and must not be used on animal samples. Refer suspected avian influenza to the State Animal Husbandry Department.

- **Instrument:** an open real-time PCR system can run veterinary kits. Check each kit's validated instruments, dye channels and passive reference before purchase.
- **Kits found on the market:** Retro Biotech (Jaipur) multiplex veterinary kits including Brucella; YouSeq canine distemper and parvovirus multiplex, plus Babesia and Ehrlichia assays; Ring Bio Flexy Pet canine babesiosis and parvovirus; Gold Standard Diagnostics INgene q Babesia; Thermo Fisher VetMAX MastiType for bovine mastitis (validated on ABI 7500 and QuantStudio 5). Indian researchers have published multiplex qPCRs for Babesia gibsoni, B. vogeli, Ehrlichia canis and Hepatozoon canis.
- **Verification per kit:** limit of detection by dilution series, repeatability, positive and negative extraction controls on every run, and comparison with a reference laboratory on 20 or more positive and negative field samples per target.
- **Extraction:** verify the extraction kit separately for EDTA blood, faeces, urine and milk.
- **Segregation:** process animal samples in separate extraction batches with distinct labels and an LIS species flag, to prevent mix-ups and cross-contamination with human work.

- **Scheduled diseases:** lumpy skin disease and other diseases in the Act's Schedule carry reporting duties. Agree a written procedure with the Animal Husbandry Department and refer positives for confirmation (Western Regional Disease Diagnostic Laboratory, Pune).
- **Biosafety:** Brucella is internationally classed as risk group 3; do not culture it. Extract all animal samples in a Class II biosafety cabinet.

4. Reference intervals

Healthy animals available	Approach (ASVCP guidelines, Friedrichs 2012; CLSI EP28-A3c)
120 or more	Establish de novo, nonparametric, with 90% confidence intervals
40 to 119	Establish by the robust method
20 to 39	Parametric if Gaussian, otherwise robust
20 (transference)	Verify a published or reference-lab interval: accept if 2 or fewer of 20 results fall outside
Fewer than 20	Do not set an interval

Start by transferring intervals for dogs and cats with 20 healthy animals per species and analyte (for example from Nagpur Veterinary College clinics and partner practices, with owner consent). Report each species, and adult and young animals where needed, separately. Livestock intervals for the dairy profile follow once a herd partner is on board.

5. Setting up the veterinary vertical

Area	Requirement
Signatory	The Indian Veterinary Council Act, 1984 (section 30) reserves veterinary practice and veterinary certificates to registered veterinary practitioners. Appoint a registered veterinary pathologist (MVSc) to sign and interpret reports.
Accreditation	Decision: no NABL accreditation for veterinary tests at launch. Run them under the existing quality management system but outside the ISO 15189 scope. Follow NABL's rules on use of its symbol (NABL 133): no NABL symbol on veterinary reports, and a line stating the tests are not covered by accreditation. Veterinary laboratories are accredited to ISO/IEC 17025; revisit this once volumes justify it.
LIS and reports	Species, breed, sex and age fields; species reference intervals; a distinct veterinary report template stating the method; no mixing of human and animal identifiers.
Waste	The Bio-Medical Waste Management Rules, 2016 apply to pathological laboratories, including animal anatomical waste.
Quality	ASVCP quality assurance guidelines (version 3.0, 2019), Westgard rules with veterinary controls, and enrolment in a veterinary external quality assessment scheme.
Samples	Supply clinics with EDTA and plain tubes, slides and cool boxes. Chilled overnight transport; separate serum before shipping; frozen EDTA plasma for ACTH.

6. Timeline and checklist

Weeks	Work
0-2	Appoint veterinary pathologist; order BA200 veterinary reagents and controls, veterinary PCR kits and extraction kits; decide on a veterinary haematology analyser; LIS changes

2-6	BA200 verification (dog, cat); haematology comparison and smear-review protocol; reference-interval transfer for chemistry and CBC
4-10	Hormone validation on ARCHITECT and MAGLUMI (T4, cortisol with dynamic tests, progesterone); choose platform per analyte
6-12	PCR kit verification: tick-fever panel, parvovirus and distemper, leptospira, bovine haemoparasites, mastitis
8	Soft launch: CBC and chemistry for partner clinics
12	Full launch with hormones and PCR; first external quality assessment round

Sources

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Search-based research, October 2026. Manufacturer capabilities and regulations change; confirm with Abbott, Snibe, Mindray, BioSystems and the State Animal Husbandry Department before acting.